Lecture 11: Node Voltage Analysis

EE201 – Circuit Theory I

M. Mert Ankaralı

Department of Electrical and Electronics Engineering
Middle East Technical University

current voltage, node voltage e supernode

Review: two ways to solve an LTI resistive circuit

Transform the circuit

- Substitution, equivalent one-ports.
- Series and parallel connections.
- Source transformation.

Works for many circuits, but not all.

Review: two ways to solve an LTI resistive circuit

Transform the circuit

- Substitution, equivalent one-ports.
- Series and parallel connections.
- Source transformation.

Works for many circuits, but not all.

Keep the original circuit

- KCL: $A\mathbf{i} = \mathbf{0}$.
- KVL: $\mathbf{v} = A^T\mathbf{e}$.
- One element law per branch.

Works for every circuit.

Review: two ways to solve an LTI resistive circuit

Transform the circuit

- Substitution, equivalent one-ports.
- Series and parallel connections.
- Source transformation.

Works for many circuits, but not all.

Keep the original circuit

- KCL: $A\mathbf{i} = \mathbf{0}$.
- KVL: $\mathbf{v} = A^T\mathbf{e}$.
- One element law per branch.

Works for every circuit.

With the node voltages $\mathbf{e}$ as the unknowns, KVL holds automatically.

Review: two ways to solve an LTI resistive circuit

Transform the circuit

- Substitution, equivalent one-ports.
- Series and parallel connections.
- Source transformation.

Works for many circuits, but not all.

Keep the original circuit

- KCL: $A\mathbf{i} = \mathbf{0}$.
- KVL: $\mathbf{v} = A^T\mathbf{e}$.
- One element law per branch.

Works for every circuit.

With the node voltages $\mathbf{e}$ as the unknowns, KVL holds automatically.

Only KCL is left to write.

Motivation

Simplifying the circuit

- So far, we solved circuits by simplifying them.

Motivation

Simplifying the circuit

- So far, we solved circuits by simplifying them.
- Simplifying fundamentally changes the circuit; that is its whole point.

Motivation

Simplifying the circuit

- So far, we solved circuits by simplifying them.
- Simplifying fundamentally changes the circuit; that is its whole point.
- To find the quantities of the original circuit, we must go back step by step.

Motivation

Simplifying the circuit

- So far, we solved circuits by simplifying them.
- Simplifying fundamentally changes the circuit; that is its whole point.
- To find the quantities of the original circuit, we must go back step by step.
- Some circuits cannot be simplified at all.

Motivation

Simplifying the circuit

- So far, we solved circuits by simplifying them.
- Simplifying fundamentally changes the circuit; that is its whole point.
- To find the quantities of the original circuit, we must go back step by step.
- Some circuits cannot be simplified at all.

Node voltage analysis

- It works on the original circuit, without changing it.

Motivation

Simplifying the circuit

- So far, we solved circuits by simplifying them.
- Simplifying fundamentally changes the circuit; that is its whole point.
- To find the quantities of the original circuit, we must go back step by step.
- Some circuits cannot be simplified at all.

Node voltage analysis

- It works on the original circuit, without changing it.
- Its unknowns are only the $n - 1$ node voltages.

Motivation

Simplifying the circuit

- So far, we solved circuits by simplifying them.
- Simplifying fundamentally changes the circuit; that is its whole point.
- To find the quantities of the original circuit, we must go back step by step.
- Some circuits cannot be simplified at all.

Node voltage analysis

- It works on the original circuit, without changing it.
- Its unknowns are only the $n - 1$ node voltages.
- From them, every branch voltage and current follows.

Motivation

Simplifying the circuit

- So far, we solved circuits by simplifying them.
- Simplifying fundamentally changes the circuit; that is its whole point.
- To find the quantities of the original circuit, we must go back step by step.
- Some circuits cannot be simplified at all.

Node voltage analysis

- It works on the original circuit, without changing it.
- Its unknowns are only the $n-1$ node voltages.
- From them, every branch voltage and current follows.
- It works for every LTI resistive circuit; voltage sources need small additions, such as the supernode.

Motivation

Simplifying the circuit

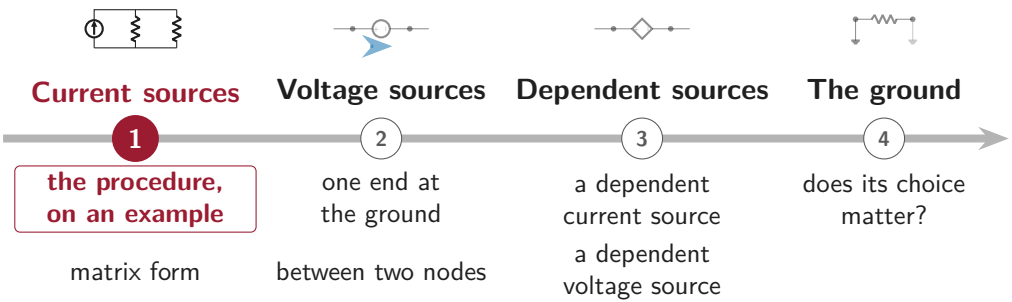
- So far, we solved circuits by simplifying them.
- Simplifying fundamentally changes the circuit; that is its whole point.
- To find the quantities of the original circuit, we must go back step by step.
- Some circuits cannot be simplified at all.

Node voltage analysis

- It works on the original circuit, without changing it.
- Its unknowns are only the $n-1$ node voltages.
- From them, every branch voltage and current follows.
- It works for every LTI resistive circuit; voltage sources need small additions, such as the supernode.

Later, mesh current analysis will do the same with mesh currents.

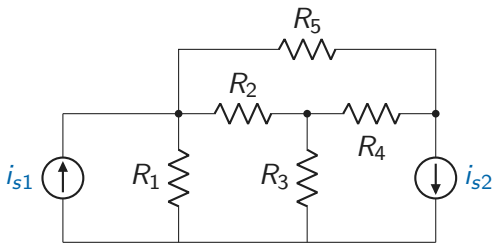
Roadmap



Each station adds a new kind of source; the last one asks about the ground.

Example 11.1: node voltage analysis step by step

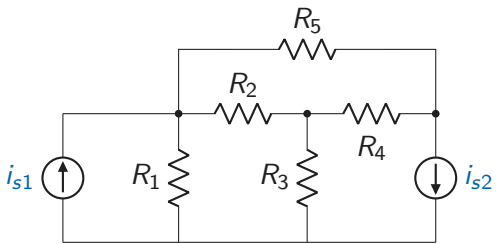
For now: the circuit has only resistors and independent current sources.



Question. How many unknowns does this circuit have?

Example 11.1: node voltage analysis step by step

For now: the circuit has only resistors and independent current sources.

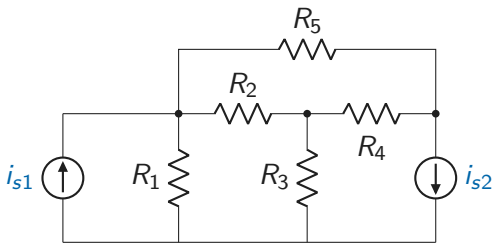


Question. How many unknowns does this circuit have?

$b = 7$ branches: 14 branch voltages and currents.

Example 11.1: node voltage analysis step by step

For now: the circuit has only resistors and independent current sources.



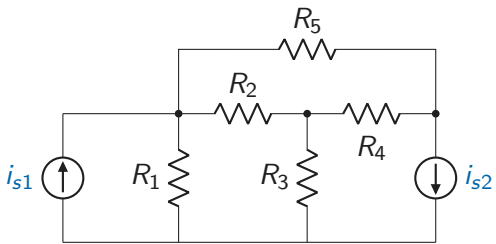
Question. How many unknowns does this circuit have?

$b = 7$ branches: 14 branch voltages and currents.

Node voltage analysis needs only 3 of them.

Example 11.1: node voltage analysis step by step

For now: the circuit has only resistors and independent current sources.



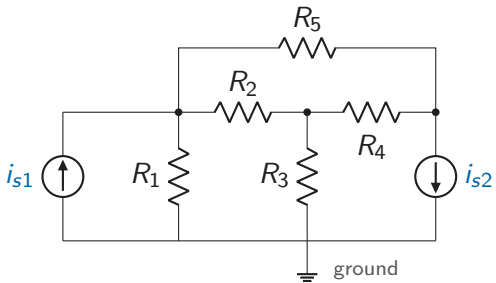
Question. How many unknowns does this circuit have?

$b = 7$ branches: 14 branch voltages and currents.

Node voltage analysis needs only 3 of them.

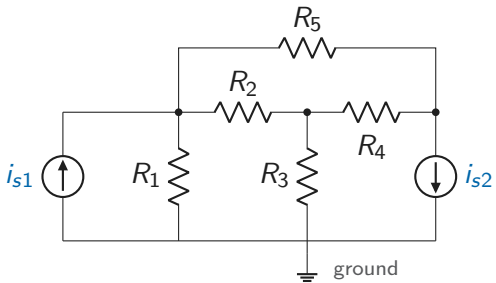
We find them in four steps, on this circuit.

Step 1: choose a reference node



Choose one node as the reference node, the ground.

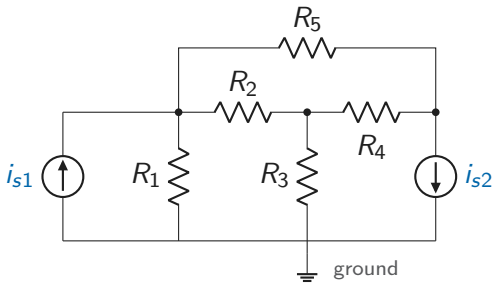
Step 1: choose a reference node



Choose one node as the reference node, the ground.

Its node voltage is 0.

Step 1: choose a reference node

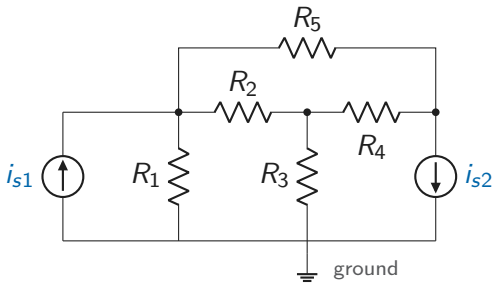


Choose one node as the reference node, the ground.

Its node voltage is 0.

Here we take the bottom node.

Step 1: choose a reference node



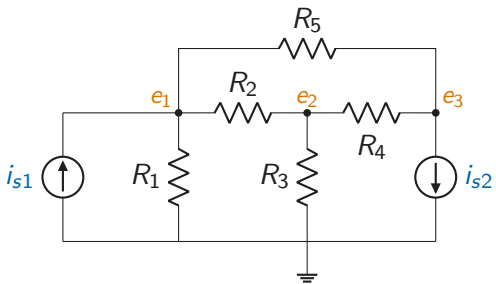
Choose one node as the reference node, the ground.

Its node voltage is 0.

Here we take the bottom node.

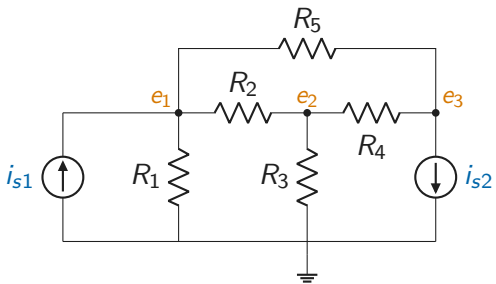
Any node can be the ground; we come back to this at the end.

Step 2: label the node voltages



Each other node gets a node voltage, measured from the ground: e_1, e_2, e_3.

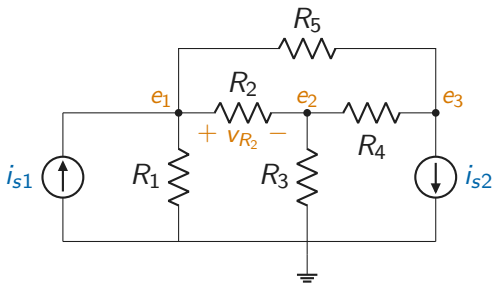
Step 2: label the node voltages



Each other node gets a node voltage, measured from the ground: e_1, e_2, e_3.

These are the unknowns.

Step 2: label the node voltages

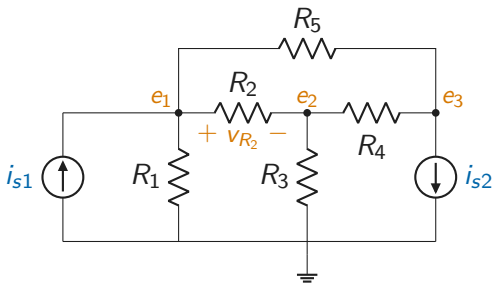


Each other node gets a node voltage, measured from the ground: e_1, e_2, e_3.

These are the unknowns.

Every branch voltage is a difference of node voltages: $v_{R_2} = e_1 - e_2$.

Step 2: label the node voltages



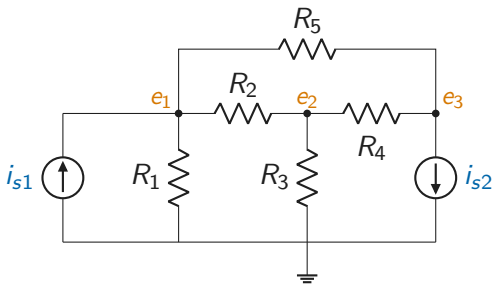
Each other node gets a node voltage, measured from the ground: e_1, e_2, e_3.

These are the unknowns.

Every branch voltage is a difference of node voltages: $v_{R_2} = e_1 - e_2$.

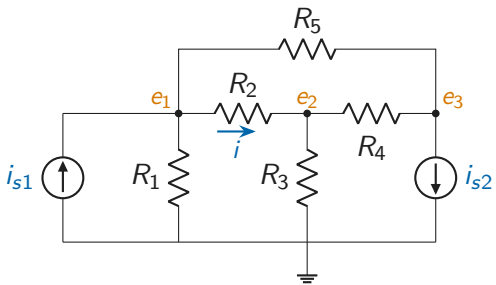
So KVL holds automatically: around R_1, R_2, R_3, $v_{R_1} = v_{R_2} + v_{R_3}$ is $e_1 = (e_1 - e_2) + e_2$.

Step 3: write the currents with node voltages



Ohm's law gives each resistor current from its two node voltages.

Step 3: write the currents with node voltages

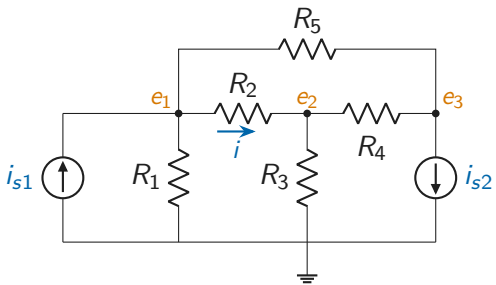


Ohm's law gives each resistor current from its two node voltages.

The current leaving node 1 through R_2:

$$i = \frac{e_1 - e_2}{R_2}$$

Step 3: write the currents with node voltages



Ohm's law gives each resistor current from its two node voltages.

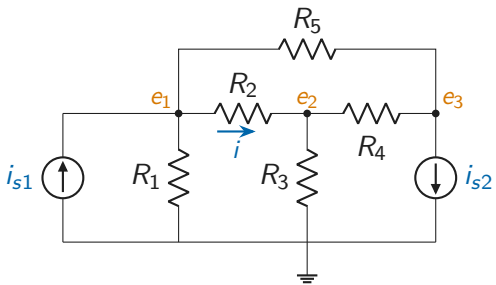
The current leaving node 1 through R_2:

$$i = \frac{e_1 - e_2}{R_2}$$

Through R_1 to the ground:

$$\frac{e_1 - 0}{R_1} = \frac{e_1}{R_1}$$

Step 3: write the currents with node voltages



Ohm's law gives each resistor current from its two node voltages.

The current leaving node 1 through R_2:

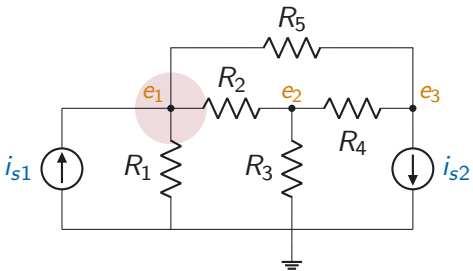
$$i = \frac{e_1 - e_2}{R_2}$$

Through R_1 to the ground:

$$\frac{e_1 - 0}{R_1} = \frac{e_1}{R_1}$$

The rule: own node voltage minus the other one, over R.

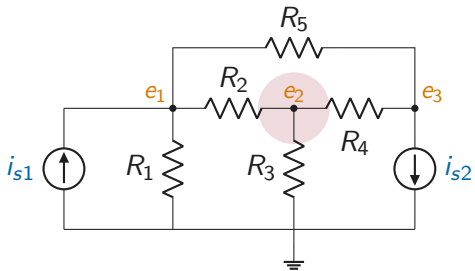
Step 4: KCL at every node except the ground



Node 1:

$$-i_{s1} + \frac{e_1}{R_1} + \frac{e_1 - e_2}{R_2} + \frac{e_1 - e_3}{R_5} = 0$$

Step 4: KCL at every node except the ground



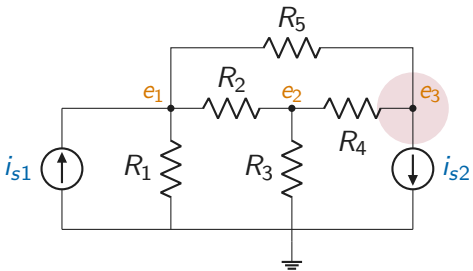
Node 1:

$$-i_{s1} + \frac{e_1}{R_1} + \frac{e_1 - e_2}{R_2} + \frac{e_1 - e_3}{R_5} = 0$$

Node 2:

$$\frac{e_2 - e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

Step 4: KCL at every node except the ground



Node 1:

$$-i_{s1} + \frac{e_1}{R_1} + \frac{e_1 - e_2}{R_2} + \frac{e_1 - e_3}{R_5} = 0$$

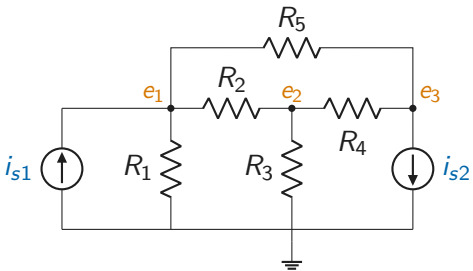
Node 2:

$$\frac{e_2 - e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

Node 3:

$$\frac{e_3 - e_2}{R_4} + \frac{e_3 - e_1}{R_5} + i_{s2} = 0$$

Step 4: KCL at every node except the ground



Node 1:

$$-i_{s1} + \frac{e_1}{R_1} + \frac{e_1 - e_2}{R_2} + \frac{e_1 - e_3}{R_5} = 0$$

Node 2:

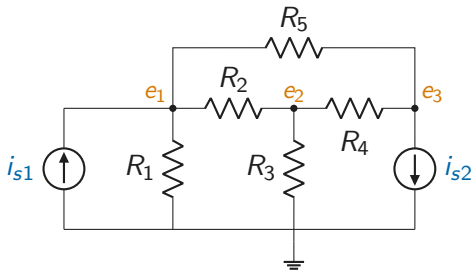
$$\frac{e_2 - e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

Node 3:

$$\frac{e_3 - e_2}{R_4} + \frac{e_3 - e_1}{R_5} + i_{s2} = 0$$

A source pointing into the node enters with a minus sign.

Step 4: KCL at every node except the ground



Node 1:

$$-i_{s1} + \frac{e_1}{R_1} + \frac{e_1 - e_2}{R_2} + \frac{e_1 - e_3}{R_5} = 0$$

Node 2:

$$\frac{e_2 - e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

Node 3:

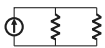
$$\frac{e_3 - e_2}{R_4} + \frac{e_3 - e_1}{R_5} + i_{s2} = 0$$

A source pointing into the node enters with a minus sign.

Three equations, three unknowns.

Then Steps 2 and 3 give every branch voltage and current.

Roadmap









Current sources | **Voltage sources** | **Dependent sources** | **The ground**

1 — 2 — 3 — 4

the procedure, on an example ✓

matrix form

one end at the ground

between two nodes

a dependent current source

a dependent voltage source

does its choice matter?

Each station adds a new kind of source; the last one asks about the ground.

Example 11.1 in matrix form

With the conductances $G_k = 1/R_k$:

$$\begin{aligned}(G_1 + G_2 + G_5)\, e_1 - G_2\, e_2 - G_5\, e_3 &= i_{s1} \\ -G_2\, e_1 + (G_2 + G_3 + G_4)\, e_2 - G_4\, e_3 &= 0 \\ -G_5\, e_1 - G_4\, e_2 + (G_4 + G_5)\, e_3 &= -i_{s2}\end{aligned}$$

Question. Write the three equations as one matrix equation.

Example 11.1 in matrix form

With the conductances $G_k = 1/R_k$:

$$
\begin{aligned}
(G_1 + G_2 + G_5)\, e_1 - G_2\, e_2 - G_5\, e_3 &= i_{s1} \\
-G_2\, e_1 + (G_2 + G_3 + G_4)\, e_2 - G_4\, e_3 &= 0 \\
-G_5\, e_1 - G_4\, e_2 + (G_4 + G_5)\, e_3 &= -i_{s2}
\end{aligned}
$$

$$
\underbrace{\begin{bmatrix} G_1 + G_2 + G_5 & -G_2 & -G_5 \\ -G_2 & G_2 + G_3 + G_4 & -G_4 \\ -G_5 & -G_4 & G_4 + G_5 \end{bmatrix}}_{Y_n}
\underbrace{\begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}}_{\mathbf{e}}
=
\underbrace{\begin{bmatrix} i_{s1} \\ 0 \\ -i_{s2} \end{bmatrix}}_{\mathbf{i}_n}
$$

Example 11.1 in matrix form

With the conductances $G_k = 1/R_k$:

$$
\begin{aligned}
(G_1 + G_2 + G_5)\, e_1 - G_2\, e_2 - G_5\, e_3 &= i_{s1} \\
-G_2\, e_1 + (G_2 + G_3 + G_4)\, e_2 - G_4\, e_3 &= 0 \\
-G_5\, e_1 - G_4\, e_2 + (G_4 + G_5)\, e_3 &= -i_{s2}
\end{aligned}
$$

$$
\underbrace{\begin{bmatrix} G_1 + G_2 + G_5 & -G_2 & -G_5 \\ -G_2 & G_2 + G_3 + G_4 & -G_4 \\ -G_5 & -G_4 & G_4 + G_5 \end{bmatrix}}_{Y_n}
\underbrace{\begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}}_{\mathbf{e}}
=
\underbrace{\begin{bmatrix} i_{s1} \\ 0 \\ -i_{s2} \end{bmatrix}}_{\mathbf{i}_n}
$$

Y_n is the node admittance matrix: $Y_n \mathbf{e} = \mathbf{i}_n$.

Example 11.1 in matrix form

With the conductances $G_k = 1/R_k$:

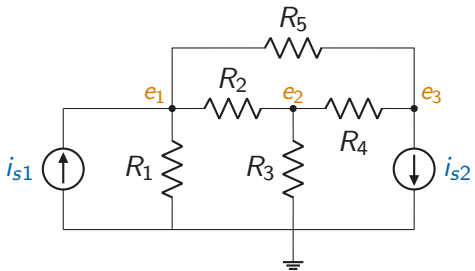
$$
\begin{aligned}
(G_1 + G_2 + G_5)\, e_1 - G_2\, e_2 - G_5\, e_3 &= i_{s1} \\
-G_2\, e_1 + (G_2 + G_3 + G_4)\, e_2 - G_4\, e_3 &= 0 \\
-G_5\, e_1 - G_4\, e_2 + (G_4 + G_5)\, e_3 &= -i_{s2}
\end{aligned}
$$

$$
\underbrace{\begin{bmatrix} G_1 + G_2 + G_5 & -G_2 & -G_5 \\ -G_2 & G_2 + G_3 + G_4 & -G_4 \\ -G_5 & -G_4 & G_4 + G_5 \end{bmatrix}}_{Y_n}
\underbrace{\begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}}_{\mathbf{e}}
=
\underbrace{\begin{bmatrix} i_{s1} \\ 0 \\ -i_{s2} \end{bmatrix}}_{\mathbf{i}_n}
$$

Y_n is the node admittance matrix: $Y_n \mathbf{e} = \mathbf{i}_n$.

$\mathbf{i}_n$: the source currents entering each node.

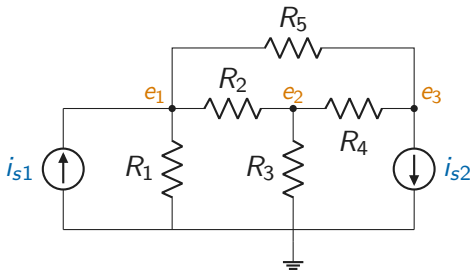
Writing Y_n directly from the circuit



$$\begin{bmatrix} G_1+G_2+G_5 & -G_2 & -G_5 \\ -G_2 & G_2+G_3+G_4 & -G_4 \\ -G_5 & -G_4 & G_4+G_5 \end{bmatrix}$$

Question. Can you read each entry of Y_n directly from the circuit?

Writing Y_n directly from the circuit

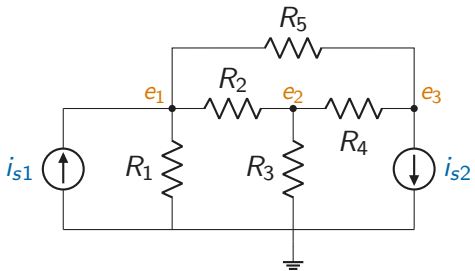


$$\begin{bmatrix} G_1+G_2+G_5 & -G_2 & -G_5 \\ -G_2 & G_2+G_3+G_4 & -G_4 \\ -G_5 & -G_4 & G_4+G_5 \end{bmatrix}$$

Question. Can you read each entry of Y_n directly from the circuit?

Diagonal (k,k): the sum of the conductances connected to node k.

Writing Y_n directly from the circuit



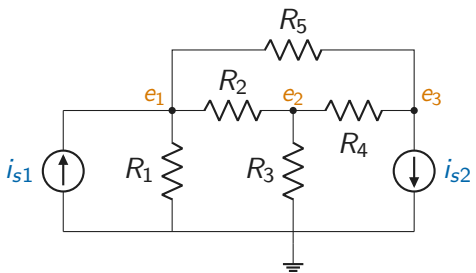
$$\begin{bmatrix} G_1+G_2+G_5 & -G_2 & -G_5 \\ -G_2 & G_2+G_3+G_4 & -G_4 \\ -G_5 & -G_4 & G_4+G_5 \end{bmatrix}$$

Question. Can you read each entry of Y_n directly from the circuit?

Diagonal (k,k): the sum of the conductances connected to node k.

Off-diagonal (k,j): minus the sum of the conductances between nodes k and j.

Writing Y_n directly from the circuit



$$\begin{bmatrix} G_1+G_2+G_5 & -G_2 & -G_5 \\ -G_2 & G_2+G_3+G_4 & -G_4 \\ -G_5 & -G_4 & G_4+G_5 \end{bmatrix}$$

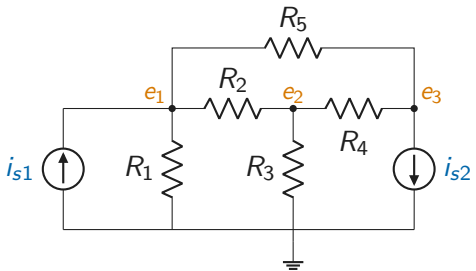
Question. Can you read each entry of Y_n directly from the circuit?

Diagonal (k,k): the sum of the conductances connected to node k.

Off-diagonal (k,j): minus the sum of the conductances between nodes k and j.

Right side k: the sum of the source currents entering node k.

Writing Y_n directly from the circuit



$$\begin{bmatrix} G_1+G_2+G_5 & -G_2 & -G_5 \\ -G_2 & G_2+G_3+G_4 & -G_4 \\ -G_5 & -G_4 & G_4+G_5 \end{bmatrix}$$

Question. Can you read each entry of Y_n directly from the circuit?

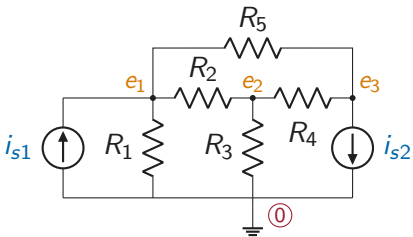
Diagonal (k,k): the sum of the conductances connected to node k.

Off-diagonal (k,j): minus the sum of the conductances between nodes k and j.

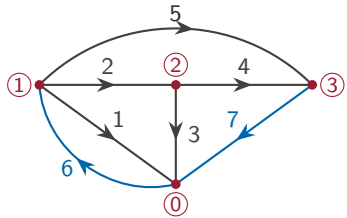
Right side k: the sum of the source currents entering node k.

So Y_n is symmetric: the entry (k,j) equals the entry (j,k).

Y_n from the graph: the reduced incidence matrix

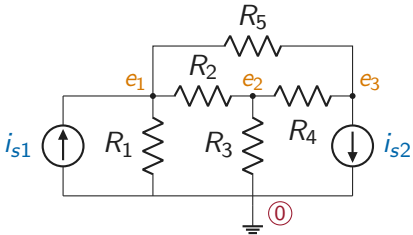


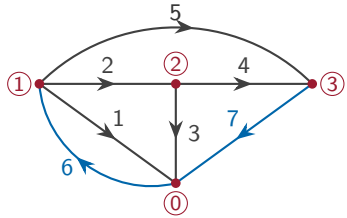
Question. Write the reduced incidence matrix A.



Branches 1–5: $R_1, \ldots, R_5$; 6, 7: i_{s1}, i_{s2}; node ⓪: the ground.

Y_n from the graph: the reduced incidence matrix





Branches 1–5: $R_1, \ldots, R_5$; 6, 7: i_{s1}, i_{s2}; node ⓪: the ground.

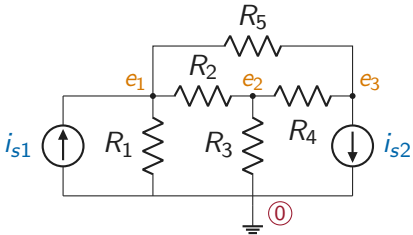
Question. Write the reduced incidence matrix A.

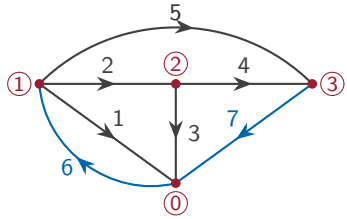
One row for each node except the ground.

Entry $+1$: the branch leaves the node; -1: it enters.

$$A = \begin{array}{c} \\ ① \\ ② \\ ③ \end{array} \begin{array}{c} \begin{array}{ccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array} \\ \left[\begin{array}{ccccccc} & & & & & & \end{array}\right] \end{array}$$

Y_n from the graph: the reduced incidence matrix





Branches 1–5: $R_1, \ldots, R_5$; 6, 7: i_{s1}, i_{s2}; node ⓪: the ground.

Question. Write the reduced incidence matrix A.

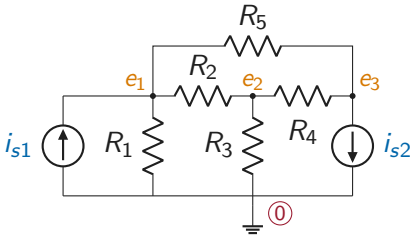
One row for each node except the ground.

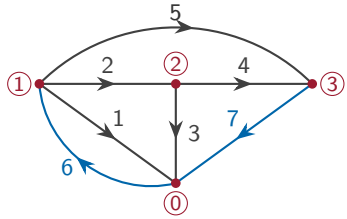
Entry $+1$: the branch leaves the node; -1: it enters.

Row k: which branches touch node k?

$$A = \begin{array}{c} \\ ① \\ ② \\ ③ \end{array} \begin{array}{c} \begin{array}{ccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array} \\ \left[\begin{array}{ccccccc} 1 & 1 & 0 & 0 & 1 & -1 & 0 \\ & & & & & & \\ & & & & & & \end{array}\right] \end{array}$$

Y_n from the graph: the reduced incidence matrix





Branches 1–5: $R_1, \ldots, R_5$; 6, 7: i_{s1}, i_{s2}; node ⓪: the ground.

Question. Write the reduced incidence matrix A.

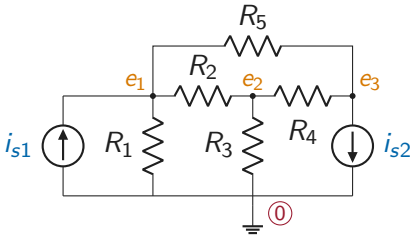
One row for each node except the ground.

Entry $+1$: the branch leaves the node; -1: it enters.

Row k: which branches touch node k?

$$A = \begin{array}{c} \\ ① \\ ② \\ ③ \end{array} \begin{array}{c} \begin{array}{ccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 1 & 0 & 0 & 0 \\ & & & & & & \end{bmatrix} \end{array}$$

Y_n from the graph: the reduced incidence matrix

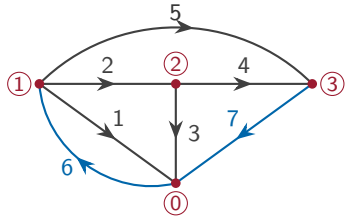


Question. Write the reduced incidence matrix A.

One row for each node except the ground.

Entry $+1$: the branch leaves the node; -1: it enters.

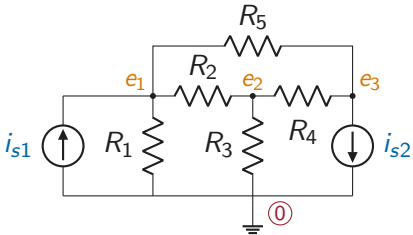
Row k: which branches touch node k?

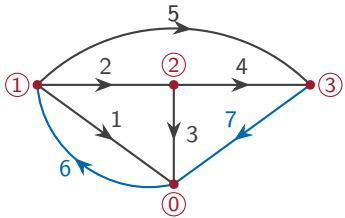


Branches 1–5: $R_1, \ldots, R_5$; 6, 7: i_{s1}, i_{s2}; node ⓪: the ground.

$$A = \begin{array}{c} \\ ① \\ ② \\ ③ \end{array} \begin{array}{c} \begin{array}{ccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & -1 & 0 & 1 \end{bmatrix} \end{array}$$

Y_n from the graph: the reduced incidence matrix





Branches 1–5: $R_1, \ldots, R_5$; 6, 7: i_{s1}, i_{s2}; node ⓪: the ground.

Question. Write the reduced incidence matrix A.

One row for each node except the ground.

Entry $+1$: the branch leaves the node; -1: it enters.

$$A = \begin{array}{c} \\ ① \\ ② \\ ③ \end{array} \overset{\begin{array}{ccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array}}{\left[\begin{array}{ccccc|cc} 1 & 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & -1 & 0 & 1 \end{array}\right]}$$

A_R: resistors A_S: sources

Y_n from the graph: KCL, KVL and Ohm's law in blocks

Number the b_R resistors first, then the b_S current sources:

$$A = [\, A_R \;\; A_S \,], \qquad \mathbf{i} = \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix}.$$

Y_n from the graph: KCL, KVL and Ohm's law in blocks

Number the b_R resistors first, then the b_S current sources:

$$A = [\, A_R \;\; A_S \,], \qquad \mathbf{i} = \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix}.$$

KCL: $A\mathbf{i} = [\, A_R \;\; A_S \,] \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix} = A_R\mathbf{i}_R + A_S\mathbf{i}_S = \mathbf{0}$

Y_n from the graph: KCL, KVL and Ohm's law in blocks

Number the b_R resistors first, then the b_S current sources:

$$A = [\, A_R \;\; A_S \,], \qquad \mathbf{i} = \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix}.$$

KCL: $A\mathbf{i} = [\, A_R \;\; A_S \,] \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix} = A_R\mathbf{i}_R + A_S\mathbf{i}_S = \mathbf{0}$

KVL: A^T splits into rows:

$$\mathbf{v} = A^T\mathbf{e}: \qquad \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix} = \begin{bmatrix} A_R^T \\ A_S^T \end{bmatrix} \mathbf{e}$$

b_R resistor rows: $\mathbf{v}_R = A_R^T\mathbf{e}$

b_S source rows: $\mathbf{v}_S = A_S^T\mathbf{e}$

Y_n from the graph: KCL, KVL and Ohm's law in blocks

Number the b_R resistors first, then the b_S current sources:

$$A = [\, A_R \;\; A_S \,], \qquad \mathbf{i} = \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix}.$$

KCL: $A\mathbf{i} = [\, A_R \;\; A_S \,] \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix} = A_R\mathbf{i}_R + A_S\mathbf{i}_S = \mathbf{0}$

KVL: A^T splits into rows:

$$\mathbf{v} = A^T\mathbf{e}: \qquad \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix} = \begin{bmatrix} A_R^T \\ A_S^T \end{bmatrix} \mathbf{e}$$

b_R resistor rows: $\mathbf{v}_R = A_R^T\mathbf{e}$

b_S source rows: $\mathbf{v}_S = A_S^T\mathbf{e}$

Only the resistor rows are needed: the source voltages follow afterwards.

Y_n from the graph: KCL, KVL and Ohm's law in blocks

Number the b_R resistors first, then the b_S current sources:

$$A = [\, A_R \;\; A_S \,], \qquad \mathbf{i} = \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix}.$$

KCL: $A\mathbf{i} = [\, A_R \;\; A_S \,] \begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix} = A_R\mathbf{i}_R + A_S\mathbf{i}_S = \mathbf{0}$

KVL: A^T splits into rows:

$$\mathbf{v} = A^T\mathbf{e}: \qquad \begin{bmatrix} \mathbf{v}_R \\ \mathbf{v}_S \end{bmatrix} = \begin{bmatrix} A_R^T \\ A_S^T \end{bmatrix} \mathbf{e}$$

b_R resistor rows: $\mathbf{v}_R = A_R^T\mathbf{e}$

b_S source rows: $\mathbf{v}_S = A_S^T\mathbf{e}$

Only the resistor rows are needed: the source voltages follow afterwards.

Resistors: $\mathbf{i}_R = G\,\mathbf{v}_R$, $G = \mathrm{diag}(G_1, \ldots, G_{b_R})$, one conductance per resistor.

Y_n from the graph: the general result

Put Ohm's law and the resistor rows of KVL into KCL:

$$A_R \mathbf{i}_R + A_S \mathbf{i}_S = A_R\, G\, \underbrace{A_R^T \mathbf{e}}_{\mathbf{v}_R} + A_S \mathbf{i}_S = \mathbf{0} \quad \Rightarrow \quad A_R\, G\, A_R^T\, \mathbf{e} = -A_S \mathbf{i}_S$$

Y_n from the graph: the general result

Put Ohm's law and the resistor rows of KVL into KCL:

$$A_R \mathbf{i}_R + A_S \mathbf{i}_S = A_R\, G\, \underbrace{A_R^T \mathbf{e}}_{\mathbf{v}_R} + A_S \mathbf{i}_S = \mathbf{0} \quad \Rightarrow \quad A_R\, G\, A_R^T\, \mathbf{e} = -A_S \mathbf{i}_S$$

$$Y_n = A_R\, G\, A_R^T, \qquad \mathbf{i}_n = -A_S \mathbf{i}_S$$

Y_n from the graph: the general result

Put Ohm's law and the resistor rows of KVL into KCL:

$$A_R \mathbf{i}_R + A_S \mathbf{i}_S = A_R\, G\, \underbrace{A_R^T \mathbf{e}}_{\mathbf{v}_R} + A_S \mathbf{i}_S = \mathbf{0} \quad \Rightarrow \quad A_R\, G\, A_R^T\, \mathbf{e} = -A_S \mathbf{i}_S$$

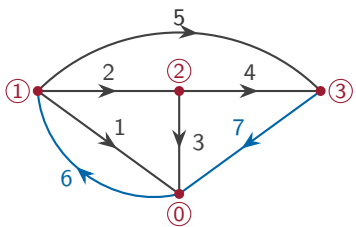
$$\boxed{Y_n = A_R\, G\, A_R^T, \qquad \mathbf{i}_n = -A_S \mathbf{i}_S}$$

$Y_n^T = A_R\, G^T A_R^T = A_R\, G\, A_R^T = Y_n$, since G is diagonal.

So Y_n is always symmetric for resistors and current sources.

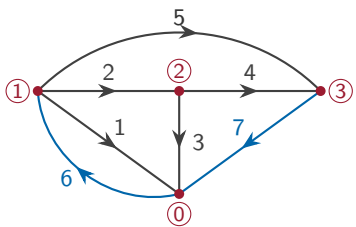
Y_n from the graph: Example 11.1, the resistor voltages

A_R^T: the resistor columns 1–5 of A, written as rows.



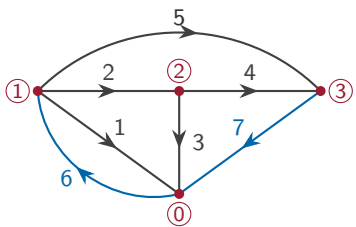
Y_n from the graph: Example 11.1, the resistor voltages

A_R^T: the resistor columns 1–5 of A, written as rows.



$$\underbrace{\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{bmatrix}}_{\mathbf{v}_R} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & -1 \end{bmatrix}}_{A_R^T} \underbrace{\begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}}_{\mathbf{e}}$$

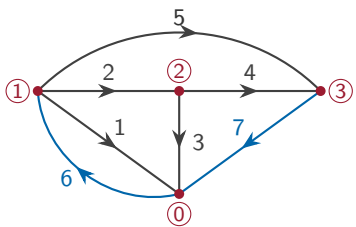
Y_n from the graph: Example 11.1, the resistor voltages



A_R^T: the resistor columns 1–5 of A, written as rows.

$$\underbrace{\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{bmatrix}}_{\mathbf{v}_R} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & -1 \end{bmatrix}}_{A_R^T} \underbrace{\begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}}_{\mathbf{e}} = \begin{bmatrix} e_1 \\ e_1 - e_2 \\ e_2 \\ e_2 - e_3 \\ e_1 - e_3 \end{bmatrix}$$

Y_n from the graph: Example 11.1, the resistor voltages

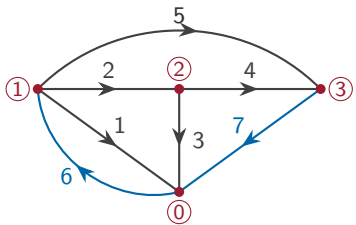


A_R^T: the resistor columns 1–5 of A, written as rows.

$$\underbrace{\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{bmatrix}}_{\mathbf{v}_R} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & -1 \end{bmatrix}}_{A_R^T} \underbrace{\begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}}_{\mathbf{e}} = \begin{bmatrix} e_1 \\ e_1 - e_2 \\ e_2 \\ e_2 - e_3 \\ e_1 - e_3 \end{bmatrix}$$

Each resistor voltage: tail node voltage minus head node voltage.

Y_n from the graph: Example 11.1, the resistor voltages



A_R^T: the resistor columns 1–5 of A, written as rows.

$$\underbrace{\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{bmatrix}}_{\mathbf{v}_R} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & -1 \end{bmatrix}}_{A_R^T} \underbrace{\begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}}_{\mathbf{e}} = \begin{bmatrix} e_1 \\ e_1 - e_2 \\ e_2 \\ e_2 - e_3 \\ e_1 - e_3 \end{bmatrix}$$

Each resistor voltage: tail node voltage minus head node voltage.

Then Ohm's law: $\mathbf{i}_R = G\,\mathbf{v}_R$, $i_k = G_k v_k$.

Y_n from the graph: Example 11.1, the matrix

$$A_R\, G = \begin{bmatrix} G_1 & G_2 & 0 & 0 & G_5 \\ 0 & -G_2 & G_3 & G_4 & 0 \\ 0 & 0 & 0 & -G_4 & -G_5 \end{bmatrix}$$

Y_n from the graph: Example 11.1, the matrix

$$A_R\, G = \begin{bmatrix} G_1 & G_2 & 0 & 0 & G_5 \\ 0 & -G_2 & G_3 & G_4 & 0 \\ 0 & 0 & 0 & -G_4 & -G_5 \end{bmatrix}$$

$$A_R\, G\, A_R^T = \begin{bmatrix} G_1 + G_2 + G_5 & -G_2 & -G_5 \\ -G_2 & G_2 + G_3 + G_4 & -G_4 \\ -G_5 & -G_4 & G_4 + G_5 \end{bmatrix}$$

Y_n from the graph: Example 11.1, the matrix

$$A_R\, G = \begin{bmatrix} G_1 & G_2 & 0 & 0 & G_5 \\ 0 & -G_2 & G_3 & G_4 & 0 \\ 0 & 0 & 0 & -G_4 & -G_5 \end{bmatrix}$$

$$A_R\, G\, A_R^T = \begin{bmatrix} G_1 + G_2 + G_5 & -G_2 & -G_5 \\ -G_2 & G_2 + G_3 + G_4 & -G_4 \\ -G_5 & -G_4 & G_4 + G_5 \end{bmatrix}$$

The same Y_n as from the three rules. ✓

Y_n from the graph: Example 11.1, the matrix

$$A_R\, G = \begin{bmatrix} G_1 & G_2 & 0 & 0 & G_5 \\ 0 & -G_2 & G_3 & G_4 & 0 \\ 0 & 0 & 0 & -G_4 & -G_5 \end{bmatrix}$$

$$A_R\, G\, A_R^T = \begin{bmatrix} G_1 + G_2 + G_5 & -G_2 & -G_5 \\ -G_2 & G_2 + G_3 + G_4 & -G_4 \\ -G_5 & -G_4 & G_4 + G_5 \end{bmatrix}$$

The same Y_n as from the three rules. ✓

$$-A_S \mathbf{i}_S = -\begin{bmatrix} -1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} i_{s1} \\ i_{s2} \end{bmatrix} = \begin{bmatrix} i_{s1} \\ 0 \\ -i_{s2} \end{bmatrix} = \mathbf{i}_n \quad ✓$$

Y_n from the graph: Example 11.1, the matrix

$$A_R\, G = \begin{bmatrix} G_1 & G_2 & 0 & 0 & G_5 \\ 0 & -G_2 & G_3 & G_4 & 0 \\ 0 & 0 & 0 & -G_4 & -G_5 \end{bmatrix}$$

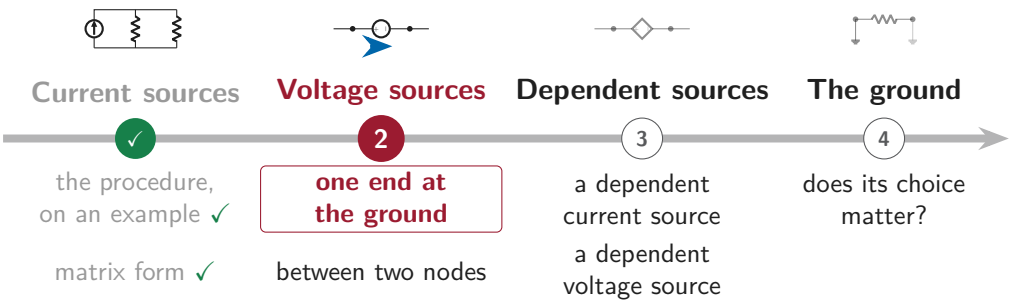
$$A_R\, G\, A_R^T = \begin{bmatrix} G_1 + G_2 + G_5 & -G_2 & -G_5 \\ -G_2 & G_2 + G_3 + G_4 & -G_4 \\ -G_5 & -G_4 & G_4 + G_5 \end{bmatrix}$$

The same Y_n as from the three rules. ✓

$$-A_S \mathbf{i}_S = -\begin{bmatrix} -1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} i_{s1} \\ i_{s2} \end{bmatrix} = \begin{bmatrix} i_{s1} \\ 0 \\ -i_{s2} \end{bmatrix} = \mathbf{i}_n \quad ✓$$

The rules are this product, read off entry by entry.

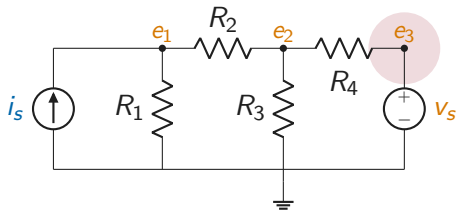
Roadmap



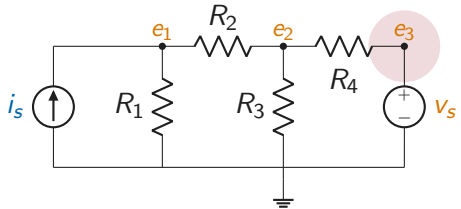
Each station adds a new kind of source; the last one asks about the ground.

Example 11.2: a voltage source to the ground

The source fixes node 3: $e_3 = v_s$ is known.



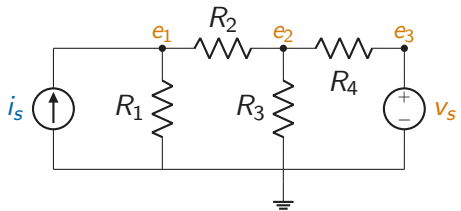
Example 11.2: a voltage source to the ground



The source fixes node 3: $e_3 = v_s$ is known.

So we do not write KCL at node 3; the unknowns are e_1, e_2.

Example 11.2: a voltage source to the ground

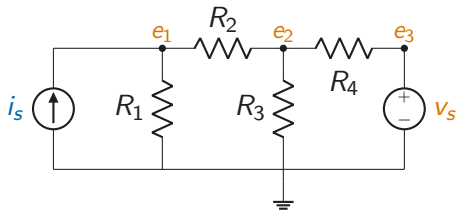


The source fixes node 3: $e_3 = v_s$ is known.

So we do not write KCL at node 3; the unknowns are e_1, e_2.

Node 1: $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

Example 11.2: a voltage source to the ground



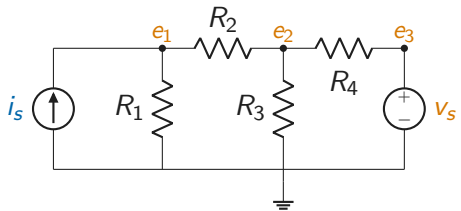
The source fixes node 3: $e_3 = v_s$ is known.

So we do not write KCL at node 3; the unknowns are e_1, e_2.

Node 1: $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

Node 2: $\dfrac{e_2 - e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - v_s}{R_4} = 0$

Example 11.2: a voltage source to the ground



The source fixes node 3: $e_3 = v_s$ is known.

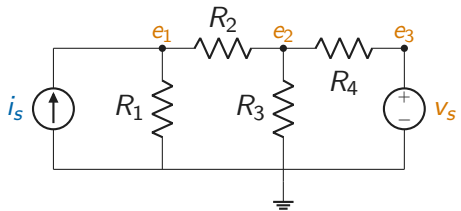
So we do not write KCL at node 3; the unknowns are e_1, e_2.

Node 1: $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

Node 2: $\dfrac{e_2 - e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - v_s}{R_4} = 0$

$$\begin{bmatrix} \frac{1}{R_1} + \frac{1}{R_2} & -\frac{1}{R_2} \\ -\frac{1}{R_2} & \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} i_s \\ v_s/R_4 \end{bmatrix}$$

Example 11.2: a voltage source to the ground



The source fixes node 3: $e_3 = v_s$ is known.

So we do not write KCL at node 3; the unknowns are e_1, e_2.

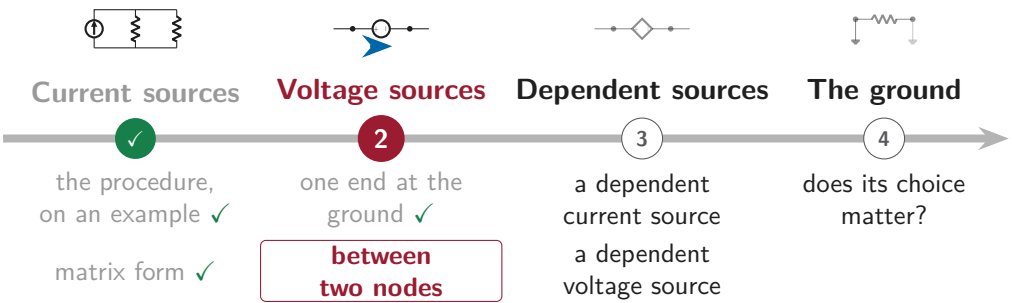
Node 1: $-i_s + \dfrac{e_1}{R_1} + \dfrac{e_1 - e_2}{R_2} = 0$

Node 2: $\dfrac{e_2 - e_1}{R_2} + \dfrac{e_2}{R_3} + \dfrac{e_2 - v_s}{R_4} = 0$

$$\begin{bmatrix} \frac{1}{R_1} + \frac{1}{R_2} & -\frac{1}{R_2} \\ -\frac{1}{R_2} & \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} i_s \\ v_s/R_4 \end{bmatrix}$$

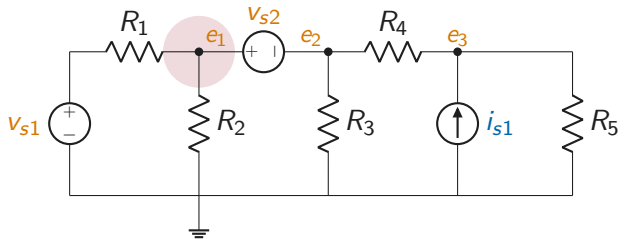
The known v_s moves to the right side.

Roadmap



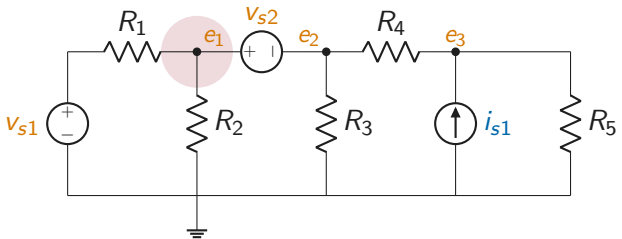
Each station adds a new kind of source; the last one asks about the ground.

Example 11.3: a voltage source between two nodes



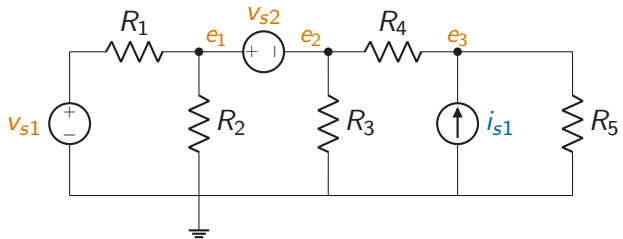
- **Question.** Write KCL at node 1: what is the current through v_{s2}?

Example 11.3: a voltage source between two nodes



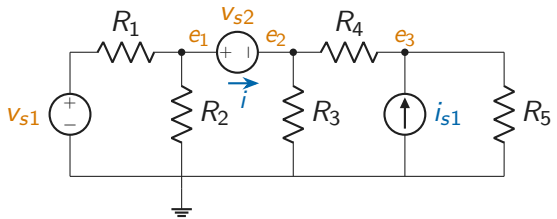
- **Question.** Write KCL at node 1: what is the current through v_{s2}?
- It cannot be written with e_1 and e_2: the law of a voltage source fixes its voltage, not its current.

Example 11.3: a voltage source between two nodes



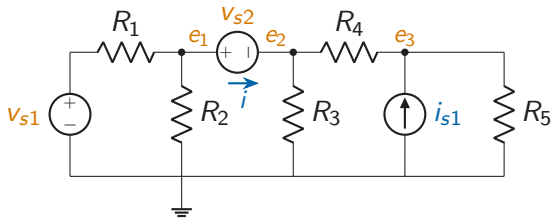
- **Question.** Write KCL at node 1: what is the current through v_{s2}?
- It cannot be written with e_1 and e_2: the law of a voltage source fixes its voltage, not its current.
- **Modified node voltage analysis** offers two ways out: Method I (its current is a new unknown) and Method II (a supernode).

Method I: the current as an unknown



The current i of v_{s2} is one more unknown.

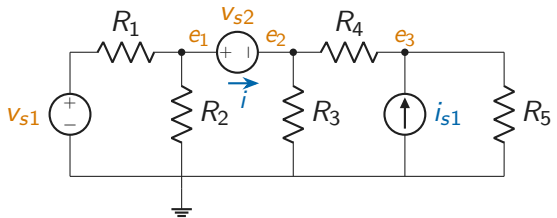
Method I: the current as an unknown



Node 1: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

The current i of v_{s2} is one more unknown.

Method I: the current as an unknown

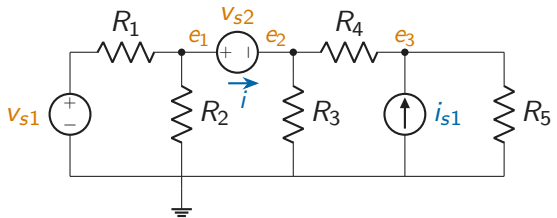


Node 1: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

Node 2: $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

The current i of v_{s2} is one more unknown.

Method I: the current as an unknown



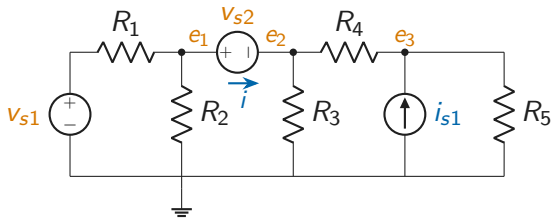
Node 1: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

Node 2: $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

The current i of v_{s2} is one more unknown.

Method I: the current as an unknown



Node 1: $\dfrac{e_1 - v_{s1}}{R_1} + \dfrac{e_1}{R_2} + i = 0$

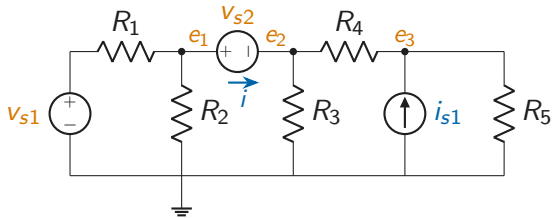
Node 2: $-i + \dfrac{e_2}{R_3} + \dfrac{e_2 - e_3}{R_4} = 0$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

The current i of v_{s2} is one more unknown.

Constraint: $e_1 - e_2 = v_{s2}$.

Method I: the current as an unknown



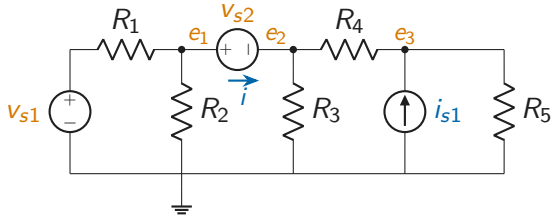
$$\begin{bmatrix} G_1+G_2 & 0 & 0 & 1 \\ 0 & G_3+G_4 & -G_4 & -1 \\ 0 & -G_4 & G_4+G_5 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ i \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ 0 \\ i_{s1} \\ v_{s2} \end{bmatrix}$$

The current i of v_{s2} is one more unknown.

Constraint: $e_1 - e_2 = v_{s2}$.

Four unknowns, four equations; the last row is the constraint.

Method I: the current as an unknown



$$\begin{bmatrix} G_1+G_2 & 0 & 0 & 1 \\ 0 & G_3+G_4 & -G_4 & -1 \\ 0 & -G_4 & G_4+G_5 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ i \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ 0 \\ i_{s1} \\ v_{s2} \end{bmatrix}$$

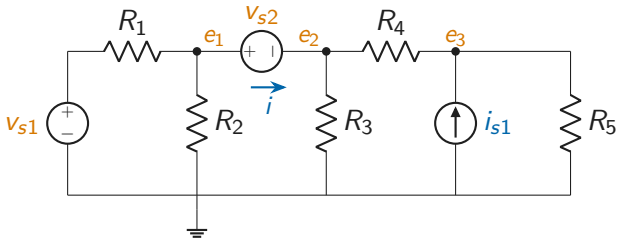
The current i of v_{s2} is one more unknown.

Constraint: $e_1 - e_2 = v_{s2}$.

Four unknowns, four equations; the last row is the constraint.

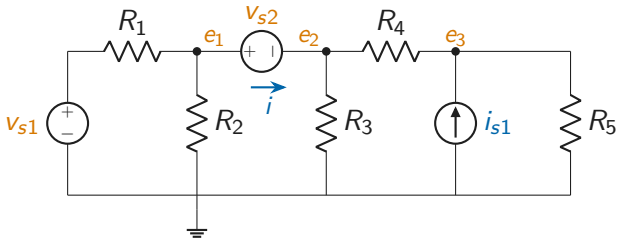
It works, but the supernode avoids the extra unknown.

Method II: the supernode



Question. Can we get rid of the unknown i?

Method II: the supernode

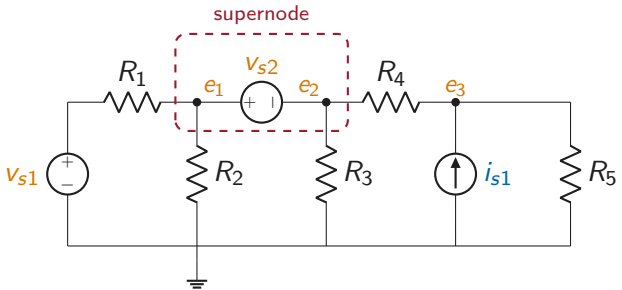


Question. Can we get rid of the unknown i?

Add the KCL equations of nodes 1 and 2.

i leaves node 1 and enters node 2, so it cancels.

Method II: the supernode



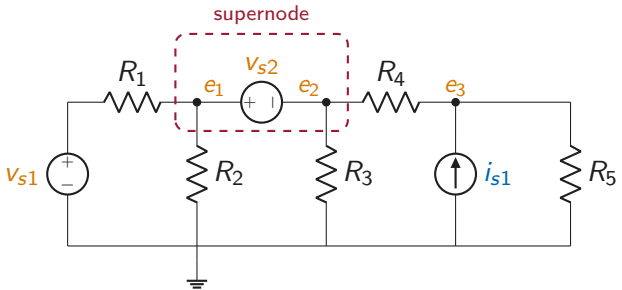
Question. Can we get rid of the unknown i?

Add the KCL equations of nodes 1 and 2.

i leaves node 1 and enters node 2, so it cancels.

This is KCL for a closed surface around both nodes and the source.

Method II: the supernode



Question. Can we get rid of the unknown i?

Add the KCL equations of nodes 1 and 2.

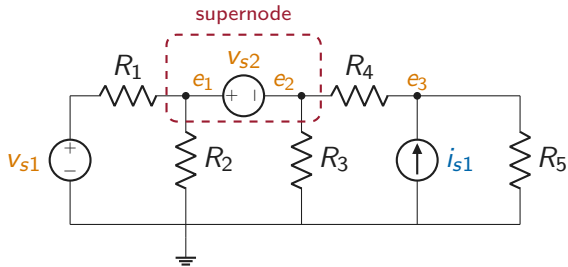
i leaves node 1 and enters node 2, so it cancels.

This is KCL for a closed surface around both nodes and the source.

Supernode

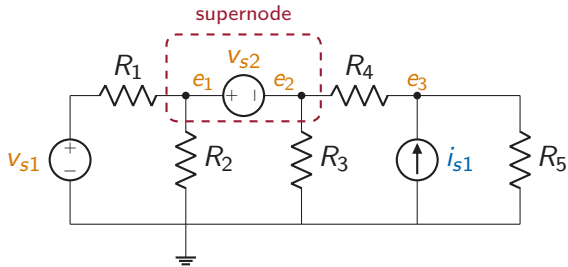
Write KCL as if the two nodes of the voltage source were a single node.

Method II: the equations



Question. Write KCL at the supernode.

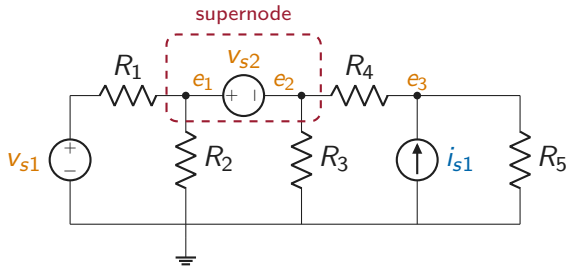
Method II: the equations



Supernode:

$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

Method II: the equations

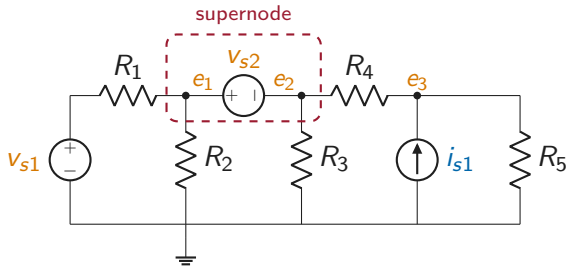


Supernode:

$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

Method II: the equations



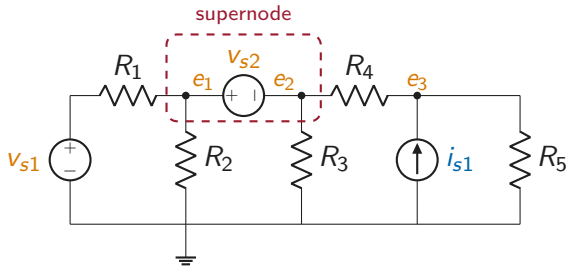
Supernode:

$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

Constraint: $e_1 - e_2 = v_{s2}$

Method II: the equations



Supernode:

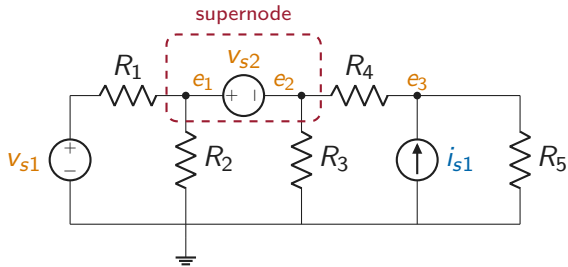
$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

Constraint: $e_1 - e_2 = v_{s2}$

$$\begin{bmatrix} G_1 + G_2 & G_3 + G_4 & -G_4 \\ 0 & -G_4 & G_4 + G_5 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ i_{s1} \\ v_{s2} \end{bmatrix}$$

Method II: the equations



Supernode:

$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

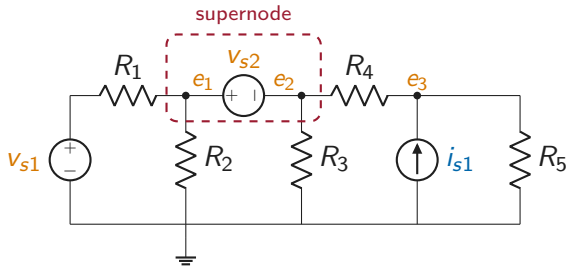
Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

Constraint: $e_1 - e_2 = v_{s2}$

$$\begin{bmatrix} G_1 + G_2 & G_3 + G_4 & -G_4 \\ 0 & -G_4 & G_4 + G_5 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ i_{s1} \\ v_{s2} \end{bmatrix}$$

Three unknowns: one fewer than Method I.

Method II: the equations



Supernode:

$$\frac{e_1 - v_{s1}}{R_1} + \frac{e_1}{R_2} + \frac{e_2}{R_3} + \frac{e_2 - e_3}{R_4} = 0$$

Node 3: $\dfrac{e_3 - e_2}{R_4} - i_{s1} + \dfrac{e_3}{R_5} = 0$

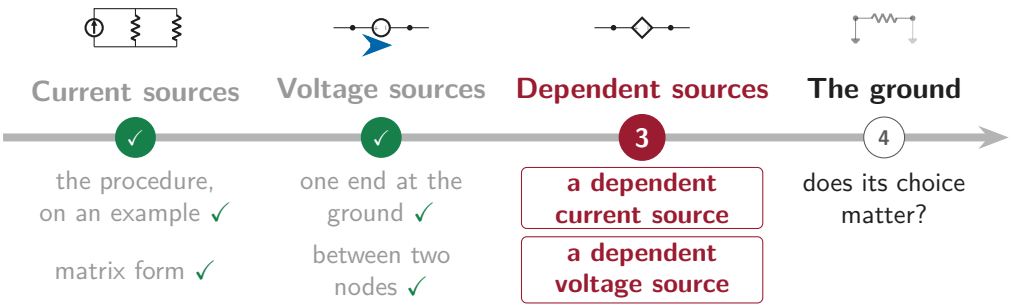
Constraint: $e_1 - e_2 = v_{s2}$

$$\begin{bmatrix} G_1 + G_2 & G_3 + G_4 & -G_4 \\ 0 & -G_4 & G_4 + G_5 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} G_1 v_{s1} \\ i_{s1} \\ v_{s2} \end{bmatrix}$$

Three unknowns: one fewer than Method I.

The matrix is not symmetric any more.

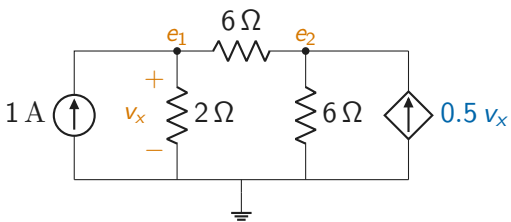
Roadmap



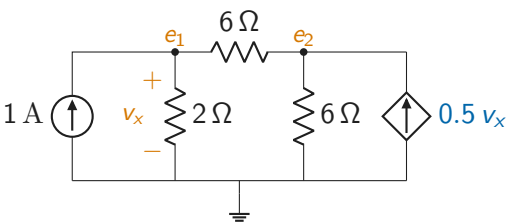
Each station adds a new kind of source; the last one asks about the ground.

Example 11.4: a dependent current source

Question. Find e_1 and e_2.



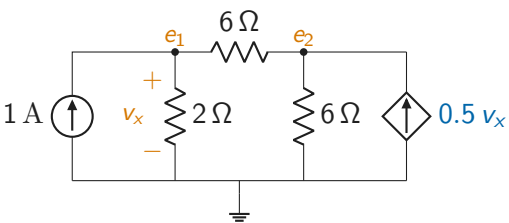
Example 11.4: a dependent current source



Question. Find e_1 and e_2.

Treat the dependent source as a current source of value $0.5\,v_x$.

Example 11.4: a dependent current source

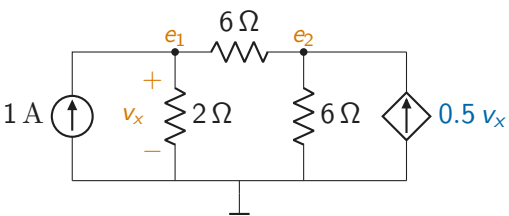


Question. Find e_1 and e_2.

Treat the dependent source as a current source of value $0.5\,v_x$.

Node 1: $-1 + \dfrac{e_1}{2} + \dfrac{e_1 - e_2}{6} = 0$

Example 11.4: a dependent current source



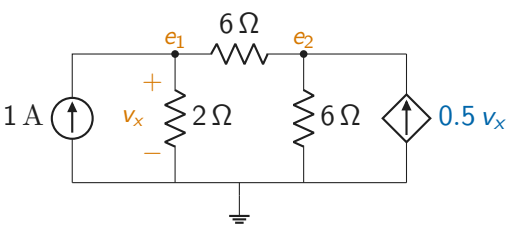
Question. Find e_1 and e_2.

Treat the dependent source as a current source of value $0.5\,v_x$.

Node 1: $$-1 + \frac{e_1}{2} + \frac{e_1 - e_2}{6} = 0$$

Node 2: $$\frac{e_2 - e_1}{6} + \frac{e_2}{6} - 0.5\,v_x = 0$$

Example 11.4: a dependent current source



Question. Find e_1 and e_2.

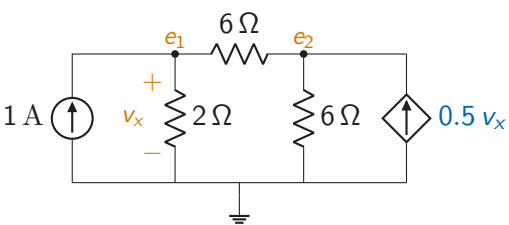
Treat the dependent source as a current source of value $0.5\,v_x$.

Node 1: $$-1 + \frac{e_1}{2} + \frac{e_1 - e_2}{6} = 0$$

Node 2: $$\frac{e_2 - e_1}{6} + \frac{e_2}{6} - 0.5\,v_x = 0$$

Question. What is v_x in node voltages?

Example 11.4: a dependent current source



Question. Find e_1 and e_2.

Treat the dependent source as a current source of value $0.5\,v_x$.

Node 1: $-1 + \dfrac{e_1}{2} + \dfrac{e_1 - e_2}{6} = 0$

Node 2: $\dfrac{e_2 - e_1}{6} + \dfrac{e_2}{6} - 0.5\,v_x = 0$

Question. What is v_x in node voltages?

The control with node voltages: $v_x = e_1$.

Example 11.4: matrix form

As if $0.5\,v_x$ were an independent source:

$$\begin{bmatrix} \frac{2}{3} & -\frac{1}{6} \\ -\frac{1}{6} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.5\,v_x \end{bmatrix}$$

Example 11.4: matrix form

As if $0.5\,v_x$ were an independent source:

$$\begin{bmatrix} \frac{2}{3} & -\frac{1}{6} \\ -\frac{1}{6} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.5\,v_x \end{bmatrix}$$

Question. Put $v_x = e_1$: which entry changes?

Example 11.4: matrix form

As if $0.5\,v_x$ were an independent source:

$$\begin{bmatrix} \frac{2}{3} & -\frac{1}{6} \\ -\frac{1}{6} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.5\,v_x \end{bmatrix}$$

Put $v_x = e_1$ and move $0.5\,e_1$ to the left side:

$$\begin{bmatrix} \frac{2}{3} & -\frac{1}{6} \\ -\frac{1}{6} - \frac{1}{2} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Example 11.4: matrix form

As if $0.5\,v_x$ were an independent source:

$$\begin{bmatrix} \frac{2}{3} & -\frac{1}{6} \\ -\frac{1}{6} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.5\,v_x \end{bmatrix}$$

Put $v_x = e_1$ and move $0.5\,e_1$ to the left side:

$$\begin{bmatrix} \frac{2}{3} & -\frac{1}{6} \\ -\frac{1}{6} - \frac{1}{2} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad \Rightarrow \qquad e_1 = 3\,\mathrm{V}, \quad e_2 = 6\,\mathrm{V}$$

Example 11.4: matrix form

As if $0.5\,v_x$ were an independent source:

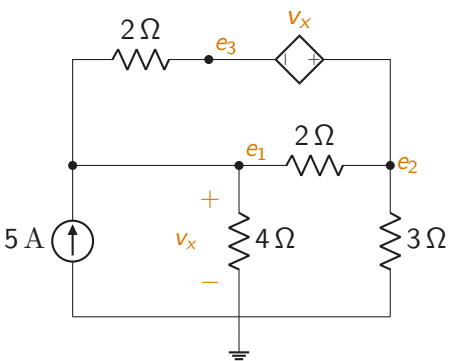
$$\begin{bmatrix} \frac{2}{3} & -\frac{1}{6} \\ -\frac{1}{6} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.5\,v_x \end{bmatrix}$$

Put $v_x = e_1$ and move $0.5\,e_1$ to the left side:

$$\begin{bmatrix} \frac{2}{3} & -\frac{1}{6} \\ -\frac{1}{6} - \frac{1}{2} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad \Rightarrow \qquad e_1 = 3\,\mathrm{V}, \quad e_2 = 6\,\mathrm{V}$$

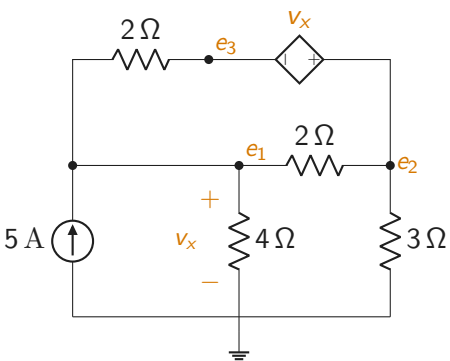
The dependent source changes one entry: Y_n is not symmetric any more.

Example 11.5: a dependent voltage source



Question. Find the power absorbed by the 3 Ω resistor.

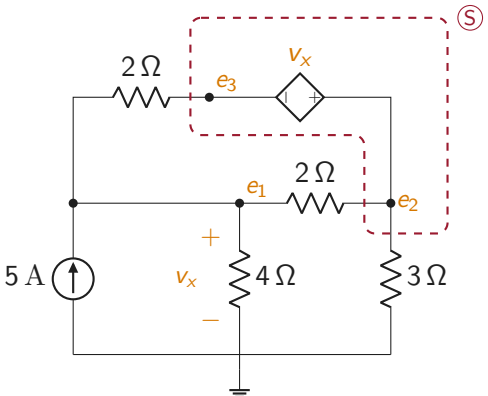
Example 11.5: a dependent voltage source



Question. Find the power absorbed by the 3 Ω resistor.

The voltage source v_x is between nodes 3 and 2; neither is the ground.

Example 11.5: a dependent voltage source

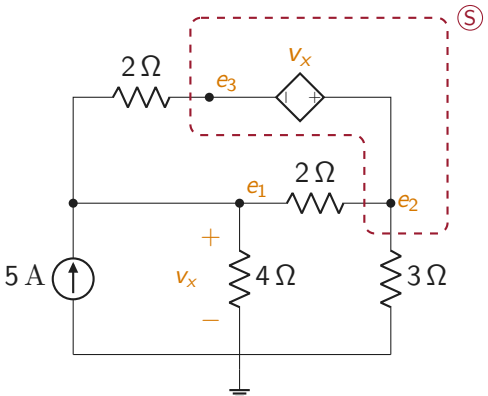


Question. Find the power absorbed by the 3 Ω resistor.

The voltage source v_x is between nodes 3 and 2; neither is the ground.

So use a supernode Ⓢ around nodes 2 and 3.

Example 11.5: a dependent voltage source



Question. Find the power absorbed by the 3 Ω resistor.

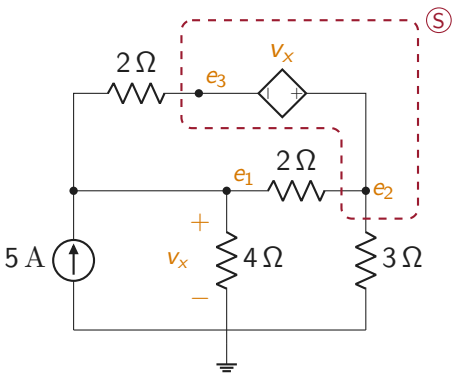
The voltage source v_x is between nodes 3 and 2; neither is the ground.

So use a supernode Ⓢ around nodes 2 and 3.

Unknowns: e_1, e_2, e_3.

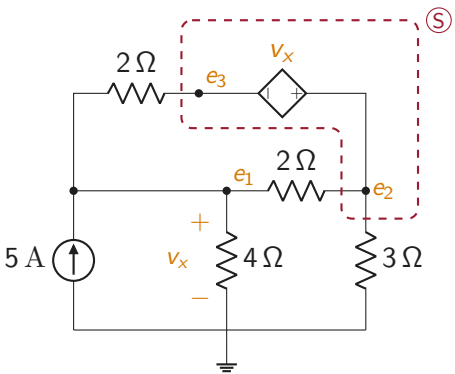
Equations: KCL at node 1, KCL at Ⓢ, and a constraint.

Example 11.5: the equations



Node 1: $$-5 + \frac{e_1 - e_3}{2} + \frac{e_1}{4} + \frac{e_1 - e_2}{2} = 0 \quad (1)$$

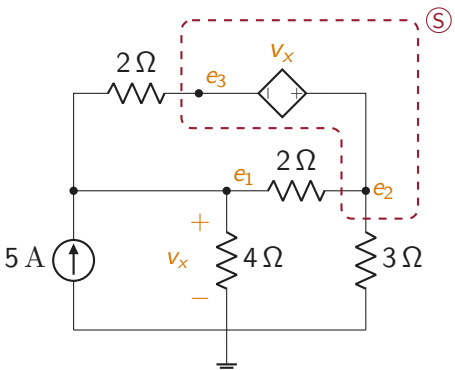
Example 11.5: the equations



Node 1: $$-5 + \frac{e_1 - e_3}{2} + \frac{e_1}{4} + \frac{e_1 - e_2}{2} = 0 \quad (1)$$

Supernode Ⓢ: $$\frac{e_3 - e_1}{2} + \frac{e_2 - e_1}{2} + \frac{e_2}{3} = 0 \quad (2)$$

Example 11.5: the equations

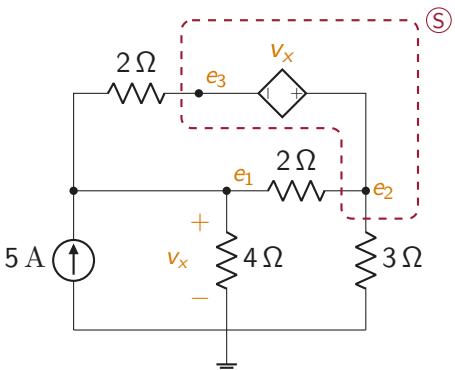


Node 1: $-5 + \dfrac{e_1 - e_3}{2} + \dfrac{e_1}{4} + \dfrac{e_1 - e_2}{2} = 0$ (1)

Supernode (S): $\dfrac{e_3 - e_1}{2} + \dfrac{e_2 - e_1}{2} + \dfrac{e_2}{3} = 0$ (2)

Question. What is the constraint?

Example 11.5: the equations



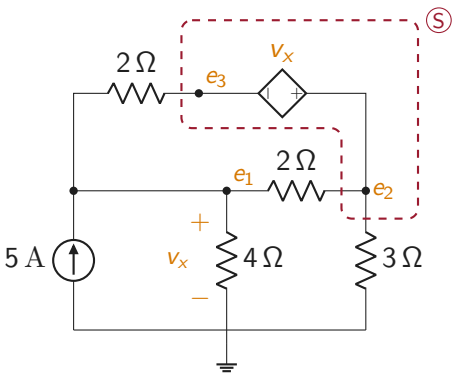
Node 1: $-5 + \dfrac{e_1 - e_3}{2} + \dfrac{e_1}{4} + \dfrac{e_1 - e_2}{2} = 0$ (1)

Supernode Ⓢ: $\dfrac{e_3 - e_1}{2} + \dfrac{e_2 - e_1}{2} + \dfrac{e_2}{3} = 0$ (2)

Question. What is the constraint?

The source: $e_2 - e_3 = v_x$.

Example 11.5: the equations



Node 1: $-5 + \dfrac{e_1 - e_3}{2} + \dfrac{e_1}{4} + \dfrac{e_1 - e_2}{2} = 0$ (1)

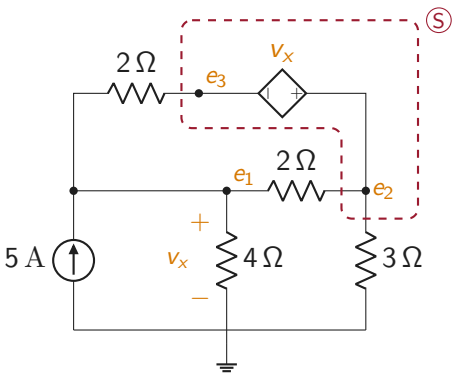
Supernode Ⓢ: $\dfrac{e_3 - e_1}{2} + \dfrac{e_2 - e_1}{2} + \dfrac{e_2}{3} = 0$ (2)

Question. What is the constraint?

The source: $e_2 - e_3 = v_x$.

Its control: $v_x = e_1$, the voltage of the 4 Ω.

Example 11.5: the equations



Node 1: $-5 + \dfrac{e_1 - e_3}{2} + \dfrac{e_1}{4} + \dfrac{e_1 - e_2}{2} = 0$ (1)

Supernode Ⓢ: $\dfrac{e_3 - e_1}{2} + \dfrac{e_2 - e_1}{2} + \dfrac{e_2}{3} = 0$ (2)

Question. What is the constraint?

The source: $e_2 - e_3 = v_x$.

Its control: $v_x = e_1$, the voltage of the 4 Ω.

Constraint: $e_2 - e_3 = e_1$ (3)

Three equations, three unknowns.

Example 11.5: matrix form

Collect the terms of (1), (2) and (3):

$$\begin{bmatrix} \frac{5}{4} & -\frac{1}{2} & -\frac{1}{2} \\ -1 & \frac{5}{6} & \frac{1}{2} \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}$$

Example 11.5: matrix form

Collect the terms of (1), (2) and (3):

$$\begin{bmatrix} \frac{5}{4} & -\frac{1}{2} & -\frac{1}{2} \\ -1 & \frac{5}{6} & \frac{1}{2} \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}$$

Rows: KCL at node 1, KCL at Ⓢ, the constraint $e_1 - e_2 + e_3 = 0$.

Example 11.5: matrix form

Collect the terms of (1), (2) and (3):

$$\begin{bmatrix} \frac{5}{4} & -\frac{1}{2} & -\frac{1}{2} \\ -1 & \frac{5}{6} & \frac{1}{2} \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}$$

Rows: KCL at node 1, KCL at Ⓢ, the constraint $e_1 - e_2 + e_3 = 0$.

Solve: $e_1 = 8\,\mathrm{V}$, $e_2 = 9\,\mathrm{V}$, $e_3 = 1\,\mathrm{V}$.

Example 11.5: matrix form

Collect the terms of (1), (2) and (3):

$$\begin{bmatrix} \frac{5}{4} & -\frac{1}{2} & -\frac{1}{2} \\ -1 & \frac{5}{6} & \frac{1}{2} \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}$$

Rows: KCL at node 1, KCL at Ⓢ, the constraint $e_1 - e_2 + e_3 = 0$.

Solve: $e_1 = 8\,\mathrm{V}$, $e_2 = 9\,\mathrm{V}$, $e_3 = 1\,\mathrm{V}$.

$$P_{3\Omega} = \frac{{e_2}^2}{3} = \frac{9^2}{3} = 27\,\mathrm{W}$$

Dependent sources

- A dependent source is handled like an independent one of the same kind.

Dependent sources

- A dependent source is handled like an independent one of the same kind.
- Its control variable (v_x here) is written with node voltages: this gives the constraint equation.

Dependent sources

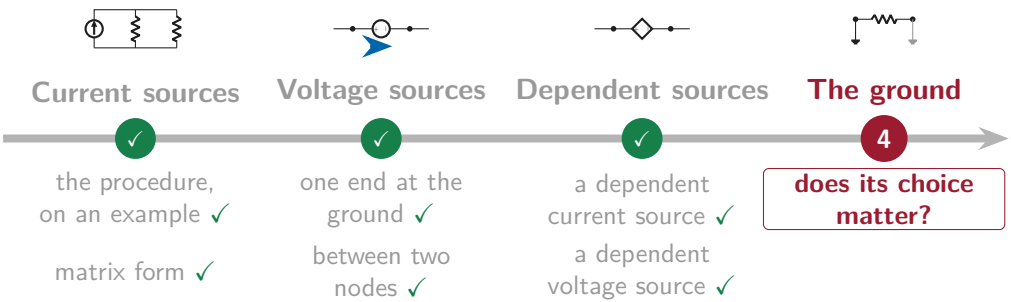
- A dependent source is handled like an independent one of the same kind.
- Its control variable (v_x here) is written with node voltages: this gives the constraint equation.

Remark

If a dependent voltage source sits between node k and the ground, do not write KCL at node k: its current is not known.

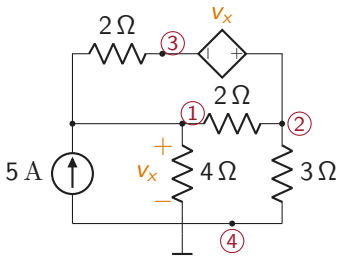
Write the constraint that the dependent source gives instead.

Roadmap

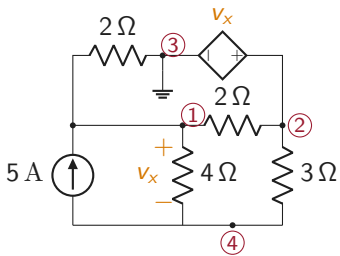


Each station adds a new kind of source; the last one asks about the ground.

Example 11.6: another ground



Example 11.5: ground at node 4

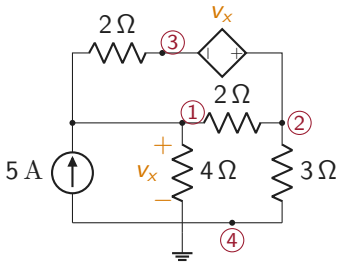


The same circuit: ground at node 3

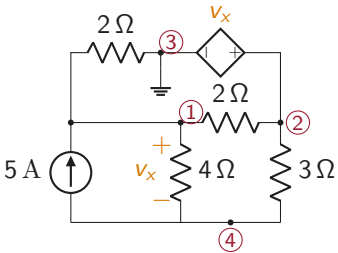
Question. Move the ground to node 3.

What happens to the node voltages?

Example 11.6: another ground



Example 11.5: ground at node 4

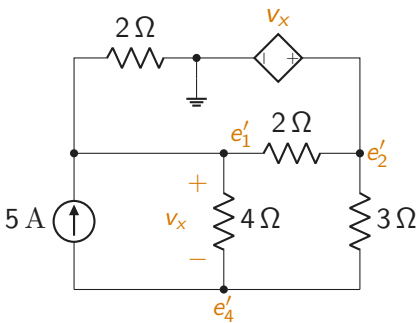


The same circuit: ground at node 3

A node voltage is measured from the ground: $e'_k = e_k - e_3 = e_k - 1\,\mathrm{V}$

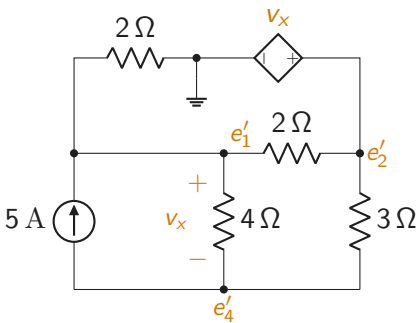
node	1	2	3	4
e_k (V)	8	9	1	0
e'_k (V)	7	8	0	−1

Example 11.6: the equations with node 3 as the ground



Question. With this ground, is a supernode needed?

Example 11.6: the equations with node 3 as the ground

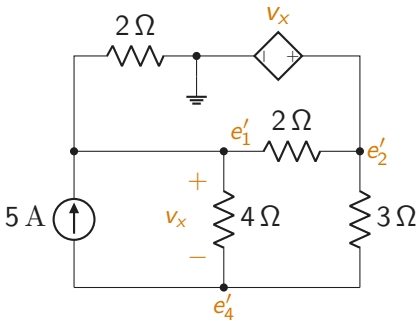


Question. With this ground, is a supernode needed?

No: v_x is now between node 2 and the ground, so $e_2' = v_x$.

Unknowns: e_1', e_2', e_4'.

Example 11.6: the equations with node 3 as the ground



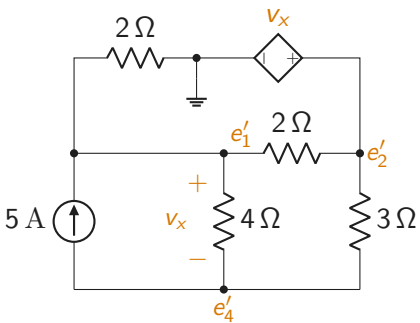
Question. With this ground, is a supernode needed?

No: v_x is now between node 2 and the ground, so $e_2' = v_x$.

Unknowns: e_1', e_2', e_4'.

Node 1: $-5 + \dfrac{e_1'}{2} + \dfrac{e_1' - e_4'}{4} + \dfrac{e_1' - e_2'}{2} = 0$

Example 11.6: the equations with node 3 as the ground



Question. With this ground, is a supernode needed?

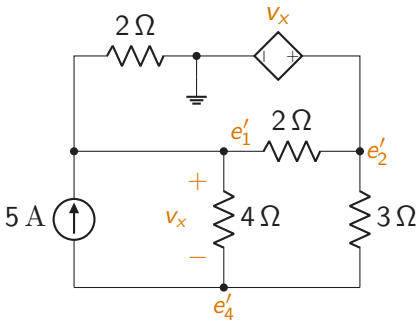
No: v_x is now between node 2 and the ground, so $e_2' = v_x$.

Unknowns: e_1', e_2', e_4'.

Node 1: $-5 + \dfrac{e_1'}{2} + \dfrac{e_1' - e_4'}{4} + \dfrac{e_1' - e_2'}{2} = 0$

Node 4: $5 + \dfrac{e_4' - e_1'}{4} + \dfrac{e_4' - e_2'}{3} = 0$

Example 11.6: the equations with node 3 as the ground



Question. With this ground, is a supernode needed?

No: v_x is now between node 2 and the ground, so $e_2' = v_x$.

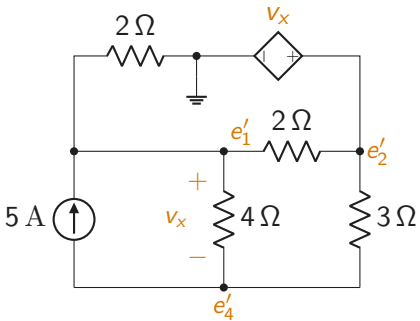
Unknowns: e_1', e_2', e_4'.

Node 1: $$-5 + \frac{e_1'}{2} + \frac{e_1' - e_4'}{4} + \frac{e_1' - e_2'}{2} = 0$$

Node 4: $$5 + \frac{e_4' - e_1'}{4} + \frac{e_4' - e_2'}{3} = 0$$

No KCL at node 2: the current of v_x is not known.

Example 11.6: the equations with node 3 as the ground



Question. With this ground, is a supernode needed?

No: v_x is now between node 2 and the ground, so $e_2' = v_x$.

Unknowns: e_1', e_2', e_4'.

Node 1: $-5 + \dfrac{e_1'}{2} + \dfrac{e_1' - e_4'}{4} + \dfrac{e_1' - e_2'}{2} = 0$

Node 4: $5 + \dfrac{e_4' - e_1'}{4} + \dfrac{e_4' - e_2'}{3} = 0$

No KCL at node 2: the current of v_x is not known.

Constraint: $e_2' = v_x = e_1' - e_4'$

Example 11.6: matrix form

Collect the terms of the two KCL equations and the constraint:

$$\begin{bmatrix} \frac{5}{4} & -\frac{1}{2} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{3} & \frac{7}{12} \\ 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} e_1' \\ e_2' \\ e_4' \end{bmatrix} = \begin{bmatrix} 5 \\ -5 \\ 0 \end{bmatrix}$$

Example 11.6: matrix form

Collect the terms of the two KCL equations and the constraint:

$$\begin{bmatrix} \frac{5}{4} & -\frac{1}{2} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{3} & \frac{7}{12} \\ 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} e_1' \\ e_2' \\ e_4' \end{bmatrix} = \begin{bmatrix} 5 \\ -5 \\ 0 \end{bmatrix}$$

Solve: $e_1' = 7\,\mathrm{V}$, $e_2' = 8\,\mathrm{V}$, $e_4' = -1\,\mathrm{V}$: each is $e_k - 1\,\mathrm{V}$, as predicted.

Example 11.6: matrix form

Collect the terms of the two KCL equations and the constraint:

$$\begin{bmatrix} \frac{5}{4} & -\frac{1}{2} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{3} & \frac{7}{12} \\ 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} e_1' \\ e_2' \\ e_4' \end{bmatrix} = \begin{bmatrix} 5 \\ -5 \\ 0 \end{bmatrix}$$

Solve: $e_1' = 7\,\mathrm{V}$, $e_2' = 8\,\mathrm{V}$, $e_4' = -1\,\mathrm{V}$: each is $e_k - 1\,\mathrm{V}$, as predicted.

Tip. Put the ground at one end of a voltage source.
Then the other node voltage is known, or a supernode is avoided.

Summary

In the circuit	What to do
resistors and current sources	KCL at every non-ground node: $Y_n\mathbf{e} = \mathbf{i}_n$, from three simple rules; Y_n is symmetric
a voltage source from the ground to node k	$e_k = v_s$ is known; no KCL at node k
a voltage source between two nodes	Method I: its current as an unknown, plus the constraint Method II: KCL at the supernode, plus the constraint
a dependent source	write its control variable with node voltages: a constraint
the choice of ground	any node; the branch voltages and currents do not change

After the node voltages: $\mathbf{v} = A^T\mathbf{e}$, then the currents from the element laws.